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Cosmic-Muon Results from the NUSEX Experiment.

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Summary. — Muon underground data collected in the NUSEX experiment at Mont Blanc have been analysed concerning the following topics: I) vertical muon intensity, II) prompt muon flux, III) muons from the direction of Cygnus X-3. Preliminary results from the analysis of more than 20 000 muons are presented and discussed.

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Introduction.

NUSEX was planned as a first-generation proton decay experiment. For this reason the apparatus has been located very deep underground (vertical depth ~ 5000 hg/cm^2 s.r.) in garage 17 in the Mt. Blanc tunnel. The uniformity

of the rock composition (granite, $Z = 10.34$, $A = 20.94$, $\varrho = 2.68$ g/cm^3) and a detailed map of the mountain showing contour lines at 10 m intervals allows us to reconstruct the slant depth in each direction with high accuracy ($\Delta h/h \sim$ $\sim 1\,\%$). Moreover, the apparatus is of simple geometry so that the acceptance is well defined and easily calculated. These properties make it suitable to perform muon physics underground.

The NUSEX detector has been described extensively in [1]. Briefly, it consists of a cube of 3.5 m side, made of 136 horizontal iron plates 1 cm thick interleaved with planes of tubes having 1 cm $\times$ 1 cm cross-section. These tubes are flushed with a gas mixture of Ar+CO_2+n-pentane and are operated in the limited streamer mode at a tension of 4000 V.

During an effective working time of 2.4 years between June '82 and December '84, 20850 single muons have been recorded satisfying the following acceptance criteria:

a) at least 10 layers of streamer chambers crossed,

b) zenith angle $\theta < 75°$.

In this way a fiducial volume is defined in which the detection efficiency is practically equal to unity. Typical errors in reconstructing tracks are $\sigma_\theta \sim 1$ mrad and $\sigma_\varphi \sim 2$ mrad. The two-track resolution is better than 2 cm.

1. – Muon intensity.

Muon intensity underground is measured in order to obtain the muon energy differential spectrum and, by calculation of the muon production in the atmosphere, the primary all-nucleon flux. The data reduction is performed in two steps as described in [2]. First the events are classified in the event matrix $\Delta\theta \times \Delta\varphi = 1° \times 1°$ (θ = zenith, φ = azimuth), and the depth and acceptance found for each bin. Starting from the event matrix the events are classified in the depth-zenith angle matrix chosen with bin width as $\Delta h \times$ $\times \Delta\theta = 200$ hg/cm$^2 \times 2°$. In this way the number of recorded events and the acceptance relative to each mean depth and mean zenith angle (in steps of 200 hg/cm^2 and 2°, respectively) are obtained.

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[1] G. BATTISTONI, E. BELLOTTI, G. BOLOGNA, P. CAMPANA, C. CASTAGNOLI, V. CHIARELLA, A. CIOCIO, D. C. CUNDY, B. D'ETTORRE PIAZZOLI, E. FIORINI, P. GALEOTTI, E. IAROCCI, C. LIGUORI, G. MANNOCCHI, G. P. MURTAS, P. NEGRI, G. NICOLETTI, P. PICCHI, M. PRICE, A. PULLIA, S. RAGAZZI, M. ROLLIER, O. SAAVEDRA, L. SATTA, L. TRASATTI and L. ZANOTTI: *Phys. Lett. B*, **133**, 454 (1983); *Nucl. Instrum. Methods*, to be published.

[2] L. BERGAMASCO, H. BILOKON, A. CASTELLINA, B. D'ETTORRE PIAZZOLI, G. MANNOCCHI, P. PICCHI and S. VERNETTO: *Nuovo Cimento C*, **6**, 569 (1983).

These data have been analysed according to the general relation

$$I_\mu(h,\theta) = I_\mu^{\pi,K}(h)\cdot\{G^{\pi,K}(h,\theta) + R(h)\,G^{p}(h,\theta)\}\,, \tag{1}$$

where $R(h) = I_\mu^{p}(h)/I_\mu^{\pi,K}(h)$ is the ratio of the vertical intensity of prompt muons to the vertical intensity of conventional ones. $G^{\pi,K}(h,\theta)$ and $G^{p}(h,\theta)$ are the angular enhancement functions which describe the dependence of the two muon components on the zenith angle θ.

From published results ([3,4]) and from preliminary analysis of the NUSEX data (next section) the contribution of prompt muons to the intensity is expected to be very low ($< 5\,\%$) even at high depth. Moreover, the major contribution to muons recorded in NUSEX at large depths comes from high zenith angle directions for which $G^{\pi,K}(h,\theta) \gg R(h)G^{p}(h,\theta)$. Accordingly the second term in (1) has been neglected in obtaining the muon vertical intensity.

The angular enhancement functions $G^{\pi,K}(h,\theta)$ have been calculated in ([3]) and found practically independent of the assumptions on the primary spectrum, cross-section behaviour at high energy and inclusive distributions. They are shown for some depths in fig. 1. It is clear that the usual approximation $G^{\pi,K}(h,\theta) \sim \sec\theta$ is acceptable only at very large depths and for $\theta < 60°$.

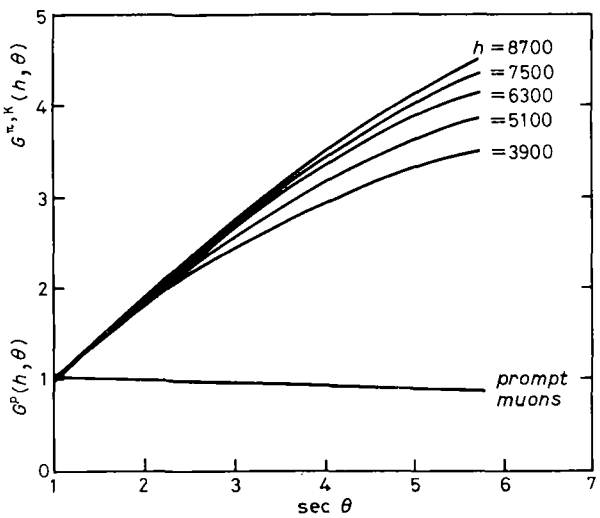


Fig. 1. – Angular enhancement of underground intensities of conventional and prompt muons.

([3]) H. BILOKON, A. CASTELLINA, B. D'ETTORRE PIAZZOLI, G. MANNOCCHI, P. PICCHI and S. VERNETTO: *Nuovo Cimento C*, **8**, 93 (1985).

([4]) I. INAZAWA, K. KOBAYAKAWA and T. KITAMURA: *A study of the contribution of prompt cosmic ray muons to total muon fluxes*, preprint KOBE-84-10.

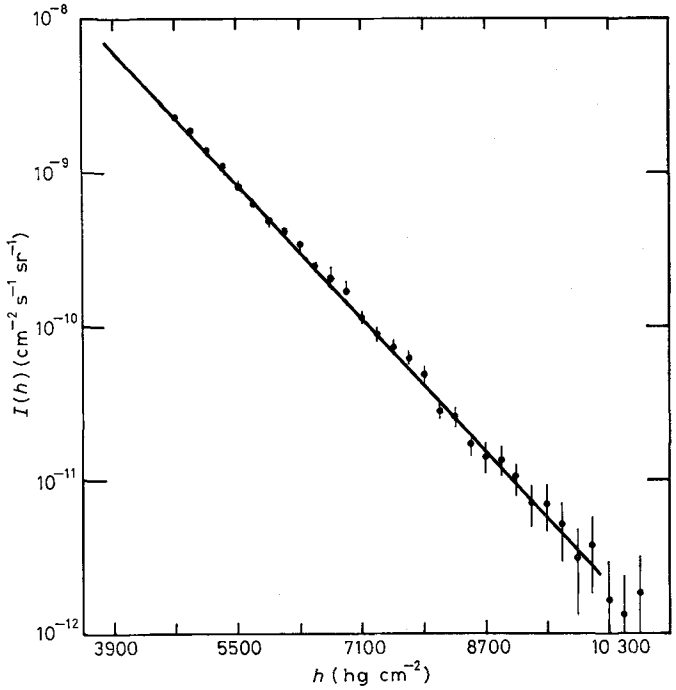


Fig. 2. – Muon intensity underground at Mt. Blanc measured in the NUSEX experiment. The exponential fit to the data (see text) is shown.

The vertical muon intensity is reported in fig. 2. The intensity *vs.* depth is very well represented in the interval (4700÷10 000) hg/cm² s.r. by the relation

$$I_\mu(h) = (7.63 \pm 0.48)\cdot 10^{-7} \exp\left[-h/(810.44 \pm 0.84)\right] \text{cm}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1}\,.$$

The agreement with the result of a previous experiment in the Mt. Blanc tunnel ([2]) is excellent. In order to obtain the muon energy spectrum a minimization procedure has been applied to the function

$$M = \sum_i \frac{1}{\sigma i^2}\left[I_\mu(h_i) - \int P(E_\mu, h_i)\,\frac{\mathrm{d}I_\mu}{\mathrm{d}E_\mu}\,\mathrm{d}E_\mu\right]^2$$

as described in ([2]). Here $P(E_\mu, h_i)$ is the survival probability function for a muon of energy E_μ at depth h_i and $\mathrm{d}I_\mu/\mathrm{d}E_\mu$ is the muon differential energy spectrum described by a power law with energy-dependent slope. A fit to the data gives

$$\frac{\mathrm{d}I_\mu}{\mathrm{d}E_\mu} = (18.24 \pm 0.12)\cdot E_\mu^{-[(3.685-0.0142)\ln(E_\mu/10^3)]}\left(\text{cm}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1}\,(\text{GeV})^{-1}\right) \quad (E_\mu \text{ in GeV}).$$

The resulting slope around 10 TeV is $\gamma = 3.71$.

The primary all-nucleon spectrum is obtained using the integral formulation for muon production and transport in the atmosphere described in (2). In this calculation the rising of the hadronic cross-sections, π and K regeneration, and energy loss in air of π, K, μ are taken into account carefully. The resulting spectrum

$$\frac{dN}{dE} = (4.8 \pm 0.2) \cdot E^{-(2.79 \pm 0.03)} \left(\text{cm}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1}\,(\text{GeV})^{-1}\right) \quad (E \text{ in GeV})$$

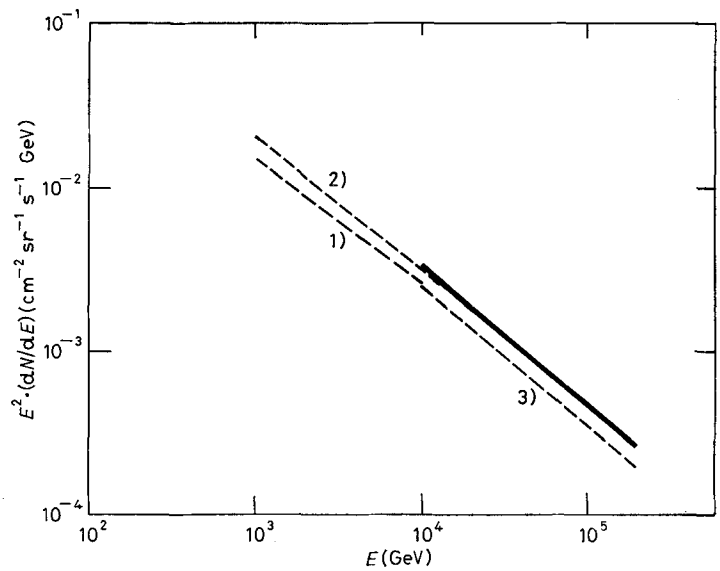


Fig. 3. – Primary all-nucleon spectrum from the NUSEX experiment (—) compared with direct Ryan (curve 1)), Grigorov (curve 2)), JACEE (curve 3), protons+helium nucleons) measurements.

up to about 10^{14} eV is compared in fig. 3 to the direct measurements of Ryan *et al.* (5), JACEE (6), Grigorov (as revised by DAS and DE (7)). The slope is higher than the direct measurements at lower energies but in good agreement with the more recent measurements from the JACEE collaboration which give $\gamma_p = 2.78 \pm 0.13$ and $\gamma_{He} = 2.83 \pm 0.20$. Adding to the proton and helium

(5) M. RYAN, J. F. ORMES and V. K. BALASUBRAHMANYAN: *Phys. Rev. Lett.*, **28**, 985, 1497(E) (1972).

(6) T. H. BURNETT, S. DUKE, M. FUKI, J. C. GREGORY, T. HAYASHI, R. HOLYNSKI, R. HUGGET, S. D. HUNTER, J. IWAI, W. V. JONES, A. JURAK, J. J. LORD, O. MIYAMURA, H. ODA, T. OGATA, T. A. PARNELL, T. SAITO, T. TABUKI, Y. TAKAHASHI, T. TOMINAGA, J. W. WATTS, B. WILCZYNSKA, R. J. WILKES, W. WOLTER and B. WOSIEK: *Phys. Rev. Lett.*, **51**, 1010 (1983).

(7) A. K. DAS and A. K. DE: *J. Phys. G*, **6**, L37 (1980).

flux measured by JACEE the nucleon contribution from the heavier nuclei groups (CNO, Mg, Fe) in the same percentage found below 1 TeV/nucleon a good consistency (within 10 %) is obtained with the NUSEX result.

2. – Prompt muons.

Experimental evidence at accelerator energies suggests that the main contribution to prompt muons should come from semi-leptonic decays of D mesons and the charmed baryon Λ_c via the reactions

$$pN \to DDX\,,$$

$$pN \to \Lambda_c DX\,.$$

Distinctive features of these muons when compared to the conventional ones are an energy dependence very close to that of the primary spectrum and an almost isotropic angular distribution. An accurate calculation of the prompt muon energy and angular distributions in the atmosphere has been performed in [8] where cross-section predictions for the two reactions from [9] have been

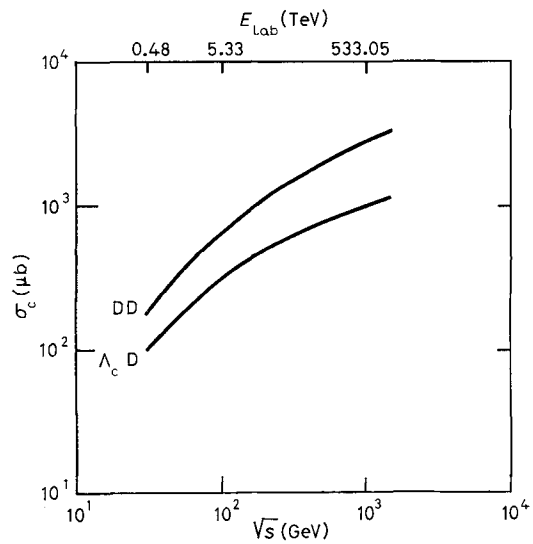


Fig. 4. – Open charm hadroproduction cross-section for the two processes pp → DDX and pp → Λ_c^+DX.

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[8] C. CASTAGNOLI, A. CASTELLINA, B. D'ETTORRE PIAZZOLI, G. MANNOCCHI, P. PICCHI and S. VERNETTO: *Nuovo Cimento A*, **82**, 78 (1984).

[9] A. Y. KHODJAMIRIAN and A. G. OGANESSIAN: EUN-652 (42) 1983, Yerevan Physics Institute.

used. The total open charm cross-section varies from 1.2 to 2.5 mb in the energy range (10÷100) TeV (fig. 4). In this calculation the inclusive distributions account for central production of D and diffractive production of Λ_c. An A and $A^{\frac{2}{3}}$ dependence on atomic number respectively has been assumed. The angular distribution of prompt muons underground is shown in fig. 1. It is not sensitive to the details of the cross-section or the shape of the primary spectrum, and is independent of the depth. This distribution is practically flat showing a small reduction of the flux at very large angles because the energy at production must increase with zenith angle due to the longer path of the muons in atmosphere.

The expected ratio between prompt and conventional muons underground is plotted in fig. 5. At a depth of 6000 hg/cm² the contribution of prompt muons is expected to be only 2 % making their detection very hard if the charm production follows the assumed energy dependence.

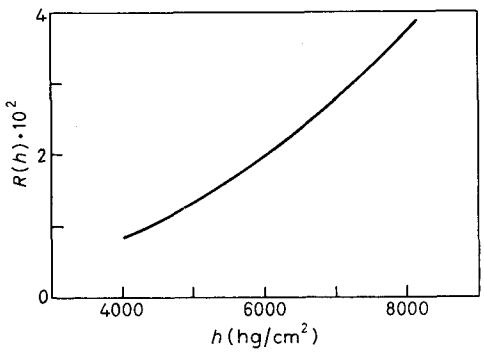


Fig. 5. – Ratio between prompt and conventional muon intensities underground.

To obtain $R(h)$ the angular enhancement function $G(h, \theta) = G^{\pi,K}(h, \theta) + R(h)G^{p}(h \cdot \theta)$ has been compared with the muon angular distribution at two mean depths, 6200 hg/cm² and 7100 hg/cm² centred on ±400 hg/cm² wide bins. The rebinning procedure applied to the data is similar to that described in [3]. This choice allows us to increase the statistics at comparable large depths, the resulting ratio $R(h)$ being in this case an average over the considered depth intervals.

The muon angular distributions at these mean depths are shown in fig. 6. The dashed line is the best fit to the data of the function $G(h, \theta)$, giving $R(6200) = (2.4 \pm 2.6)\,\%$ and $R(7100) = (4.2^{+3.6}_{-3.8})\,\%$. The experimental result is affected by very large errors because of the small angular range covered by the measurement. Unfortunately the mountain shape produces a strong correlation between zenith angle and depth which does not allow us to disentangle the dependence of the muon intensity on these parameters. The present indication is to exclude a large increase beyond the prediction of the open charm hadro-

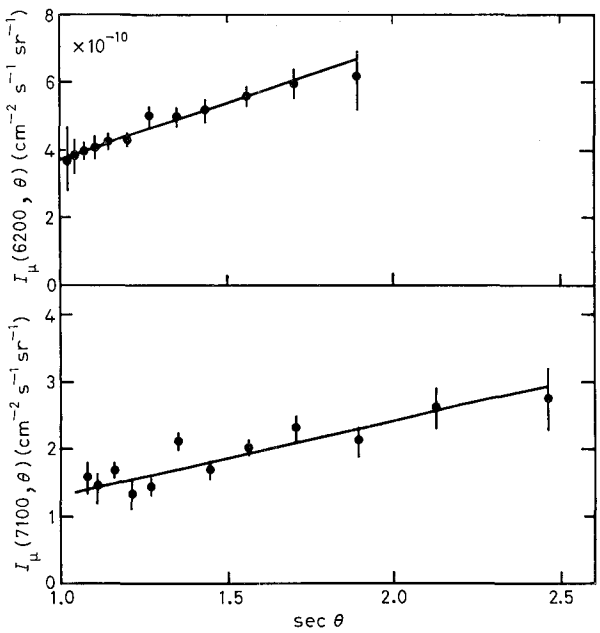


Fig. 6. – Experimental muon angular distributions at two mean depths, 6200 and 7100 hg/cm^2 s.r.

production cross-section in the energy range (20÷100) TeV involved in this measurement. An increase of the statistics and a global fit to all data could improve the sensitivity and pick up the small signal of the prompt muons.

3. – Muons from the Cygnus X-3 direction.

A program to study narrow-angle anisotropies ([10]) was started in the first months of 1984 and afterwards tuned to Cygnus X-3 by adding the phase analysis. The suggestion came from the puzzling result in the Kiel experiment ([11]) of a large muon content in showers from the Cygnus X-3 direction, higher than expected in the case of a photon origin. Ground-based observations by means of Čerenkov light detectors and air shower arrays ([12]) have established that Cygnus X-3 is a powerful source of high-energy particles observed at two different phases of about 0.2÷0.3 and about 0.6÷0.8. The energy threshold in Čerenkov experiments is typically 10^{12} eV, while scintillator apparatus sample

([10]) S. C. BARROWES: *Astrophys. J.*, **177**, 45 (1972).
([11]) M. SAMORSKI and W. STAMM: *Astrophys. J. Lett.*, **268**, L17 (1983).
([12]) J. LLOID-EVANS, R. N. COY, A. LAMBERT, J. LAPIKENS, M. PATEL, R. J. O. REID and A. A. WATSON: *Nature (London)*, **305**, 784 (1983).

air showers at larger energies $> 10^{15}$ eV. The nature of the primaries is not identified, but the observed signal is attributed to high-energy photons because i) the particle must be relatively stable with $\tau \cdot \gamma \geqslant 35\,000$ light years, ii) must be electrically neutral in order to keep the source direction over a distance $\geqslant 10$ kpc and iii) for « continuity » reasons because emission with the source periodicity has been observed at infra-red, X-ray and γ-ray energies.

Data from the period between 1 June 1982 and 1 February 1985 have been analysed. During this time 21 700 single muons with zenity angle up to 75° have been recorded satisfying the acceptance criterion. With this angular cut Cygnus X-3 is observed for 64 % of the time. In a window $10° \times 10°$ centred around the source [$\delta = 40.9$, $\alpha = 307.9$] 151 events are found. The time of each event is corrected to refer to the barycentre of the solar system and then folded modulo $4.8h$ using the Van der Klis and Bonnet-Bidaud quadratic ephemeris obtained from a fit to X-ray data

$$T_0 = JD\ 2\,440\,949.8986\,,$$

$$p\ = 0.199\,683\,0\ d\,,$$

$$\dot{p}\ = 1.18 \cdot 10^{-9}\,,$$

32 events are found in the phase range $0.7 \div 0.8$ giving an excess of 19 ± 5 over the expected background of 13 events per bin calculated by looking in the same declination band $\pm 30°$ right ascension off source, fig. 7. The phase histogram

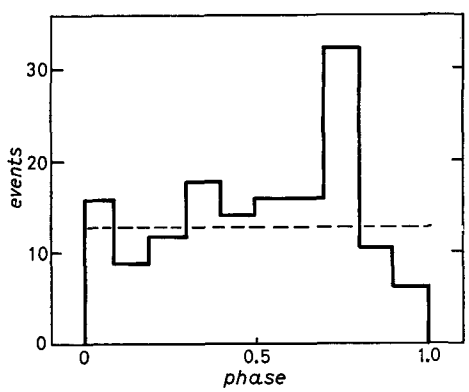


Fig. 7. – Phase distribution for muons coming from the region $10° \times 10°$ around the direction of Cygnus X-3. The expected background is shown by a dashed line.

of the 3900 off-source muons is shown in fig. 8. The flatness indicates the absence of effects due to rock thickness and that the exposure is independent of the Cygnus X-3 phase. In fact, since the Cygnus X-3 orbital period is almost exactly $\frac{1}{5}$ of the sidereal day, a strong phase/acceptance/depth correlation could

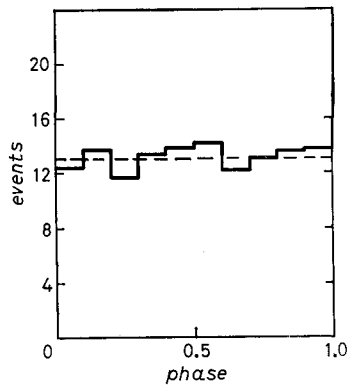


Fig. 8. – Phase distribution for off-source muons normalized to $10° \times 10°$ window. The mean value is shown by a dashed line.

be generated by noncontinuous measurements extended over a short time. Fortunately a given phase precesses over the detector 2.1 times/year producing an average effect in the NUSEX measurement extended for 2.66 years with an effective working time $\sim 90\,\%$. The exposure integrated over the running time is shown in fig. 9 as a function of the depth and is practically independent of the phase. This assures that no biases were introduced in data taking and that the probability of detecting a muon does not depend on phase.

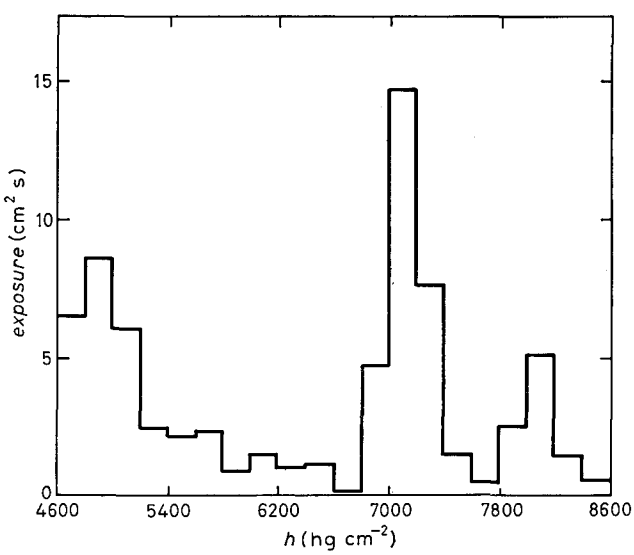


Fig. 9. – Exposure for Cygnus X-3 integrated over the running time and averaged over the total period.

The statistical significance of the excess has been estimated by a χ^2-test for uniformity ($\chi^2 = 29.1$ for 9 d.o.f. corresponding to $P(> \chi^2) = 6.10^{-4}$) and by the probability of a Poissonian fluctuation in any bin ($P = 9.10^{-5}$). The relative likelihood test ([13]) for enhanced flux against the hypothesis of a fluctuation gives a confidence level of 99.94 %.

The angular distribution of the « in phase » events is spread around Cygnus X-3 more than the expectation, fig. 10. In fact, from the tracking resolution,

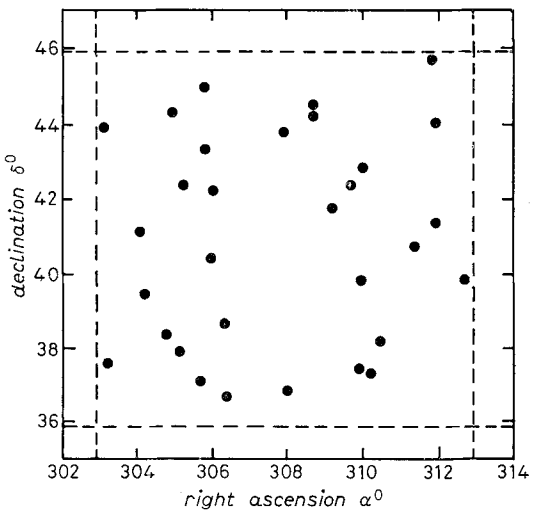


Fig. 10. – Angular distribution for on-source muons in the phase interval 0.7 ÷ 0.8.

effects of multiple scattering in rock and geomagnetic deflection ($\sim 0.6°$ as estimated from a study of the parallelism of muon pair events) and misalignement of the apparatus ($< 0.1°$) the overall errors in declination and right ascension are estimated to be $\sim 0.5°$ and $1.0°$, respectively. No easy interpretation of this effect has been found. In order to obtain the intensity underground of the observed muon signal the depth distributions of the 119 « off phase » and that of the 32 « in phase » events have been considered separately. The « off phase » events follow the distribution expected for atmospheric muons. The « in phase » events are distributed according to fig. 11. In the same figure the expected depth distribution of the 13 background events is shown by a dashed line. The subtraction of this background allows us to obtain the depth distribution of the signal. This distribution does not follow the distribution in depth of the exposure, thus ruling out the hypothesis of muons generated by neutrinos in the surrounding rock. In particular, only 1 event is found in

([13]) D. HEARN: *Nucl. Instrum. Methods*, **70**, 200 (1969).

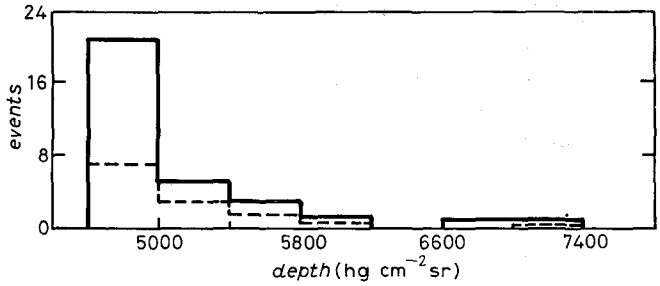


Fig. 11. – Depth distribution for the 32 « in phase » events. The expected background is shown by a dashed line, the experimental data by a full line.

the depth region around 7000 hg/cm² corresponding to the maximum of the exposure.

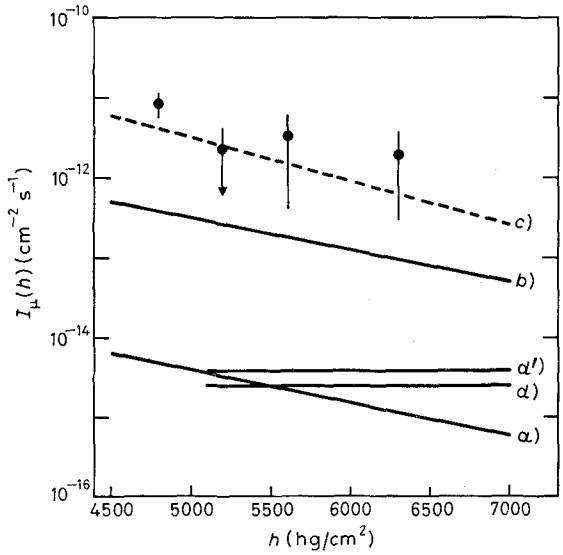


Fig. 12. – Underground intensity of muons from the direction of Cygnus X-3 compared with the expectations (curves *a*), *b*), *d*), *d'*); see text). The atmospheric background per bin in a window $10° \times 10°$ (curve *c*)) is shown.

Using the exposure distribution the intensity (averaged over the Cygnus X-3 period) at four mean depths has been obtained, fig. 12. The atmospheric background per bin in a window $10° \times 10°$ is shown for comparison (dashed line). In the same figure the expected muon intensity associated with the excess air showers from the direction of Cygnus X-3 is reported for both the hypothesis of a photonic (curve *a*) and hadronic (curve *b*) nature of the primaries. In

these calculations ([14]) a primary spectrum $7 \cdot 10^{-7} E^{-2}$ particles/cm^2·s has been used. The photoproduction cross-section on nucleons was assumed to increase logarithmically with energy. According to VMD the normalized photon nucleon inclusive distributions for π production are assumed to be similar to that of the π-nucleon interaction.

The expected muon flux induced by neutrinos from Cygnus X-3, as calculated in ([15,16]), is represented in fig. 12 by curves *d*) and *d'*). The two calculations are in agreement when the results are normalized to the same luminosity $L = 10^{39}$ erg/s.

There is not a direct explanation of the muon excess found in the direction of Cygnus X-3. Conventional interactions of photons or hadrons are unable to explain the large flux detected in NUSEX. The depth distribution of the excess muons rules out the production by conventional neutrinos interacting in the rock. Moreover, the spread of these events around the source is at variance with the resolution of the apparatus.

4. – Conclusion.

The NUSEX apparatus, although of small size, is producing significant results in cosmic-ray physics. The muon intensity underground has been measured with a good accuracy which will be improved by further running for two years. A preliminary analysis of multimuon events, already published ([17]), gives the indication that the iron fraction in the primary spectrum does not increase noticeably at high energies $> 10^{14}$ eV. A reliable limit on the open charm hadroproduction cross-section in the 100 TeV region could in future be obtained. The exciting result of a high muon flux from the direction of Cygnus X-3 has to be confirmed with more data but already poses the question of the nature of this « cosmic accelerator » and of the existence of new particles or interactions.

([14]) H. BILOKON, A. CASTELLINA, B. D'ETTORRE PIAZZOLI, G. MANNOCCHI, E. MERONI, P. PICCHI and S. VERNETTO: *Lett. Nuovo Cimento*, **44**, 491 (1985).

([15]) V. S. BEREZINSKY, C. CASTAGNOLI and P. GALEOTTI: *Nuovo Cimento C*, **8**, 185 (1985).

([16]) T. K. GAISSER and T. STANEV: *Phys. Rev. Lett.*, **54**, 2265 (1985).

([17]) G. BATTISTONI, E. BELLOTTI, C. BLOISE, G. BOLOGNA, P. CAMPANA, C. CASTAGNOLI, A. CASTELLINA, V. CHIARELLA, A. CIOCIO, D. C. CUNDY, B. D'ETTORRE PIAZZOLI, E. FIORINI, P. GALEOTTI, E. IAROCCI, C. LIGUORI, G. MANNOCCHI, G. P. MURTAS, P. NEGRI, G. NICOLETTI, P. PICCHI, M. PRICE, A. PULLIA, S. RAGAZZI, M. ROLLIER, O. SAAVEDRA, L. SATTA, S. VERNETTO and L. ZANOTTI: *Nuovo Cimento C*, **8**, 76 (1985).

● RIASSUNTO

Gli eventi di muoni rivelati sottoterra nell'esperimento NUSEX al Monte Bianco sono stati analizzati per studiare I) l'intensità verticale dei muoni, II) il flusso di muoni pronti, III) l'eccesso di muoni dalla direzione di Cygnus X-3. In questo lavoro sono presentati e discussi i risultati preliminari ottenuti dal'lanalisi di piú di 20.000 eventi.

Данные по космическим мюонам из эксперимента NUSEX.

Резюме (*). - - Был проведен анализ данных, собранных в экспеименте NUSEX на Мон Блане. Анализировались следующие вопросы: 1) интенсивность вертикальных мюонов, 2) поток мгновенных (быстрых) мюонов, 3) мюоны, движущиеся в направлении от Лебедь X-3. Приводятся и обсуждаются предварительные результаты анализа для более, чем 20000 мюонов.

(*) *Переведено редакцией.*